

Early acquisition of S-specific Tfh clonotypes after SARS-CoV-2 vaccination is associated with the longevity of anti-S antibodies

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Abstract

SARS-CoV-2 vaccines have been used worldwide to combat COVID-19 pandemic. To elucidate the factors that determine the longevity of spike (S)-specific antibodies, we traced the characteristics of S-specific T cell clonotypes together with their epitopes and anti-S antibody titers before and after BNT162b2 vaccination over time. T cell receptor (TCR) $\alpha\beta$ sequences and mRNA expression of the S-responded T cells were investigated using single-cell TCR- and RNA-sequencing. Highly expanded 199 TCR clonotypes upon stimulation with S peptide pools were reconstituted into a reporter T cell line for the determination of epitopes and restricting HLAs. Among them, we could determine 78 S epitopes, most of which were conserved in variants of concern (VOCs). In donors exhibiting sustained anti-S antibody titers (designated as “sustainers”), S-reactive T cell clonotypes detected immediately after 2nd vaccination polarized to follicular helper T (Tfh) cells, which was less obvious in “decliners”. Even before vaccination, S-reactive CD4⁺ T cell clonotypes did exist, most of which cross-reacted with environmental or symbiotic bacteria. However, these clonotypes contracted after vaccination. Conversely, S-reactive clonotypes dominated after vaccination were undetectable in pre-vaccinated T cell pool, suggesting that highly-responding S-reactive T cells were established by vaccination from rare clonotypes. These results suggest that *de novo* acquisition of memory Tfh cells upon vaccination contributes to the longevity of anti-S antibody titers.

eLife assessment

This study presents a **valuable** finding on the key factors of T cell responses associated with durable antibody responses following COVID-19 mRNA vaccinations. The data were collected with **solid** methods and approaches, but the interpretation of the conclusion may be biased due to the experimental design. If confirmed, it may have a great impact on future COVID-19 vaccine design. However, some of the evidence supporting the claims of the authors is **incomplete** with a small size of the samples, unproven validity of using the expanded T cells, and over-interpretation on the sequence homology data.

Introduction

The pandemic COVID-19, caused by the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), has expanded worldwide [1]. Many types of vaccines have been developed or in basic and clinical phases to combat infection and deterioration of COVID-19 [2,3]. Among them, messenger ribonucleic acid (mRNA) vaccines, BNT162b2/Comirnaty and mRNA-1273/Spikevax, have been approved with over 90% efficacy at 2 months post-2nd dose vaccination [4,5], and widely used. Pathogen-specific antibodies are one of the most efficient components to prevent infection. Yet, mRNA vaccine-induced serum antibody titer is known to be waning over 6 months [6,7]. Accordingly, the effectiveness of the vaccines decreases over time, and thus multiple doses and repeated boosters are necessary [8].

The production and sustainability of spike (S)-specific antibody could be related to multiple factors, especially in the case of humans [7,9]. Among them, the characteristics of SARS-CoV-2-specific T cells is critically involved in the affinity and longevity of the antibodies [10–12]. Elucidation of the key factors of T cell responses that contribute to the durable immune responses induced by vaccination would provide valuable information for the vaccine development in the future. However, the relationship between antibody sustainability and the types of antigen-specific T cells has not been investigated in a clonotype resolution.

Recent studies reported that S-reactive T cells pre-existed before exposure to SARS-CoV-2 [13–17]. Common cold human coronaviruses (HCoV) including strains 229E, NL63, OC43, and HKU1 are considered major cross-reactive antigens that primed these pre-existing T cells [15,18–20], while bacterial cross-reactive antigens were also reported [21,22]. However, the functional relevance of cross-reactive T cells during infection or vaccination is still in debate. In this study, both humoral and cellular immune responses were evaluated at 3, 6 and 24 weeks after BNT162b2/Comirnaty vaccination. S-specific T cells before and after vaccination were analyzed on clonotype level using single cell-based T cell receptor (TCR) and RNA sequencing to determine their characteristics and epitopes in antibody sustainers and decliners. These analyses suggest the importance of early acquisition of S-specific T_{fh} cells in the longevity of antibodies.

Results

SARS-CoV-2 mRNA vaccine elicits transient humoral immunity

Blood samples were collected from a total of 43 individuals (Table 1) who had no SARS-CoV-2 infection history when they received two doses of SARS-CoV-2 mRNA vaccine BNT162b2. Samples were taken before and after the vaccination (Fig. 1A). Consistent with the previous report [4],

most participants exhibited more severe side effects after 2nd dose of vaccination than 1st dose locally (**Table 2**) and systemically (**Table 3**). At 3 weeks, anti-S IgG antibody titer increased in most participants. At 6 weeks, anti-S antibody titer was at its peak. S antibody titer gradually decreased over 24 weeks (**Fig. 1B**). The antibody titer was reduced by 56.8% on average. Donors of different genders or age groups showed no significant difference in anti-S antibody titer (**Fig. S1**). The neutralization activity of the post-vaccinated sera showed similar tendency with the anti-S antibody titer during the study period (**Fig. 1C**). The above results indicate that the mRNA vaccine effectively activated humoral immune responses in healthy individuals, but decreased by 24 weeks over time as reported [6,7].

Antibody sustainers had highly expanded S-reactive Tfh clonotypes

To address the role of T cells in maintaining the antibody titer, we analyzed the S-responsive T cells in the post-vaccination samples from 8 donors, among whom 4 donors showed relatively sustained anti-S antibody titer during 6 weeks to 24 weeks (reduction < 30%) (sustainers, donors #8, #25, #27 and #28), while the other 4 donors showed largely declined anti-S antibody titer (reduction > 80%) (decliners, donors #4, #13, #15 and #17) (**Fig. 2A** and **Fig. S2A**). The possibility of SARS-CoV-2 infection of sustainers was ruled out by analyzing anti-nucleocapsid protein (N) antibody titer in the sera samples at 24 weeks (**Fig. S2B**). Antibody sustainability did not correlate with bulk T cell responses to S protein, such as IFN ψ production (**Fig. S2C**).

To enrich the S-reactive T cells, we labeled the peripheral blood mononuclear cells (PBMCs) with a cell proliferation tracer and stimulated the PBMCs with an S peptide pool for 10 days. Proliferated T cells were sorted and analyzed by single-cell TCR- and RNA-sequencing (scTCR/RNA-seq). Clustering analysis was done with pooled samples of 3 time points from 8 donors, and various T cell subtypes were identified (**Fig. 2B**). We found that, overall, the S-reactive T cells did not skew to any particular T cell subset (**Fig. 2B**). However, by grouping the cells from decliners and sustainers separately, we found difference in the frequency of the cells within the circled population (**Fig. 2C**). These cells showed high Tfh signature scores and expressed characteristic genes of Tfh cells (**Fig. 2D**), suggesting that they might be circulating Tfh cells (cTfh) considering they were isolated from PBMCs. This tendency became more pronounced when we selected highly expanded (top 16) clonotypes in each donor (**Fig. 2E**). In sustainers, S-specific Tfh clusters appeared from 6 weeks (**Fig. 2F**), suggesting that vaccine-induced Tfh cells were established immediately after 2nd vaccination.

Identification of dominant S epitopes recognized by vaccine-induced T cell clonotypes

To elucidate the epitopes of the highly expanded clonotypes, we reconstituted their TCRs into a T cell hybridoma lacking endogenous TCRs and having an NFAT-GFP reporter gene. These cell lines were stimulated with S peptides using transformed autologous B cells as antigen-presenting cells (APCs). The epitopes of 53 out of 128 reconstituted clonotypes were successfully determined (**Fig. 3**, **Table 4**, **Figs. S3A–S3D**). Epitopes of expanded Tfh cells were not limited in any particular region of S protein (**Fig. 3**). About 72% of these epitopes conserved in Delta and Omicron variants (**Tables 4** and **5**). Within the rest of 28% of epitopes which were mutated in variants of concerns (VOCs), although some mutated epitopes located in the receptor-binding domain (RBD) of VOCs lost antigenicity, recognition of most epitopes outside the RBD region was maintained or rather increased in the variants (**Table 5** and **Figs. S3E and S3F**). These results suggest that the majority of S-reactive clonotypes after vaccination can respond to antibody-escaping VOCs.

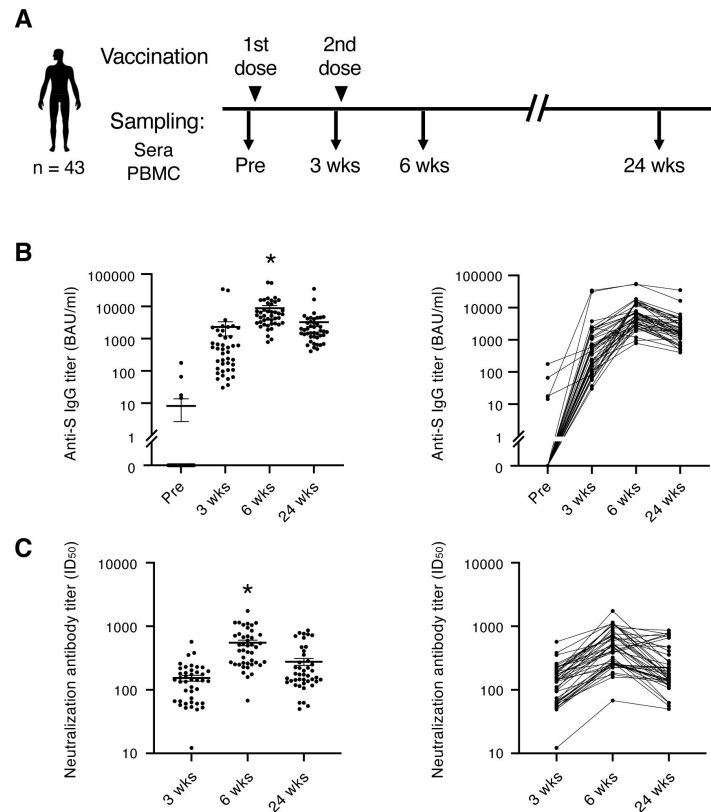


Fig. 1.

SARS-CoV-2 mRNA vaccine elicits transient humoral immunity.

(A) Vaccination and sampling timeline of blood donors in this study. (B) Anti-S IgG titer of serum samples was determined by ELISA. Mean ± SEM (left) and individual data (right) are shown. *, $P < 0.05$ vs. Pre, 3 wks, 24 wks, respectively. (C) Neutralization activity (ID₅₀) of serum samples was determined by pseudo-virus assay. Mean ± SEM (left) and individual data (right) are shown. *, $P < 0.05$ vs. Pre, 3 wks, 24 wks, respectively. Wks, weeks.

	Percentage (number)
Total number	100% (43)
Age group	
20-39	39.5% (17)
40-49	30.2% (13)
50-59	25.6% (11)
60-69	4.7% (2)
Gender	
Male	60.5% (26)
Female	39.5% (17)

Table 1.

Demographic data of the participants.

	Percentage (number)
Swelling (injection site)	
After 1st dose	27.9% (12)
After 2nd dose	51.2% (22)
Sore/pain (injection site)	
After 1st dose	88.4% (38)
After 2nd dose	86.0% (37)
Warmth (injection site)	
After 1st dose	32.6% (14)
After 2nd dose	41.9% (18)

Table 2.

Demographic data of the reported clinical adverse effects (at injection site)

	Percentage (number)
Fever	
After 1st dose	
Mild (37.5°C ≥)	2.3% (1)
Severe (≥ 38.0°C)	0% (0)
After 2nd dose	
Mild (37.5°C ≥)	25.6% (11)
Severe (≥ 38.0°C)	23.3% (10)
Fatigue	
After 1st dose	
Mild	18.6% (8)
Severe	0% (0)
After 2nd dose	
Mild	67.4% (29)
Severe	18.6% (8)
Headache	
After 1st dose	
Mild	7.0% (3)
Severe	0% (0)
After 2nd dose	
Mild	32.6% (14)
Severe	7.0% (3)
Chill	
After 1st dose	
Mild	4.7% (2)
Severe	0% (0)
After 2nd dose	
Mild	23.3% (10)
Severe	9.3% (4)
Nausea	
After 1st dose	
Mild	0% (0)
Severe	0% (0)
After 2nd dose	
Mild	4.7% (2)
Severe	0% (0)
Diarrhea	
After 1st dose	
Mild	0% (0)
Severe	0% (0)
After 2nd dose	
Mild	0% (0)
Severe	0% (0)
Muscle pain	
After 1st dose	
Mild	48.8% (21)
Severe	0% (0)
After 2nd dose	
Mild	55.8% (24)
Severe	4.7% (2)
Joint pain	
After 1st dose	
Mild	4.7% (2)
Severe	0% (0)
After 2nd dose	
Mild	25.6% (11)
Severe	4.7% (2)

Table 3.

Demographic data of the reported clinical adverse effects (systemic symptoms)

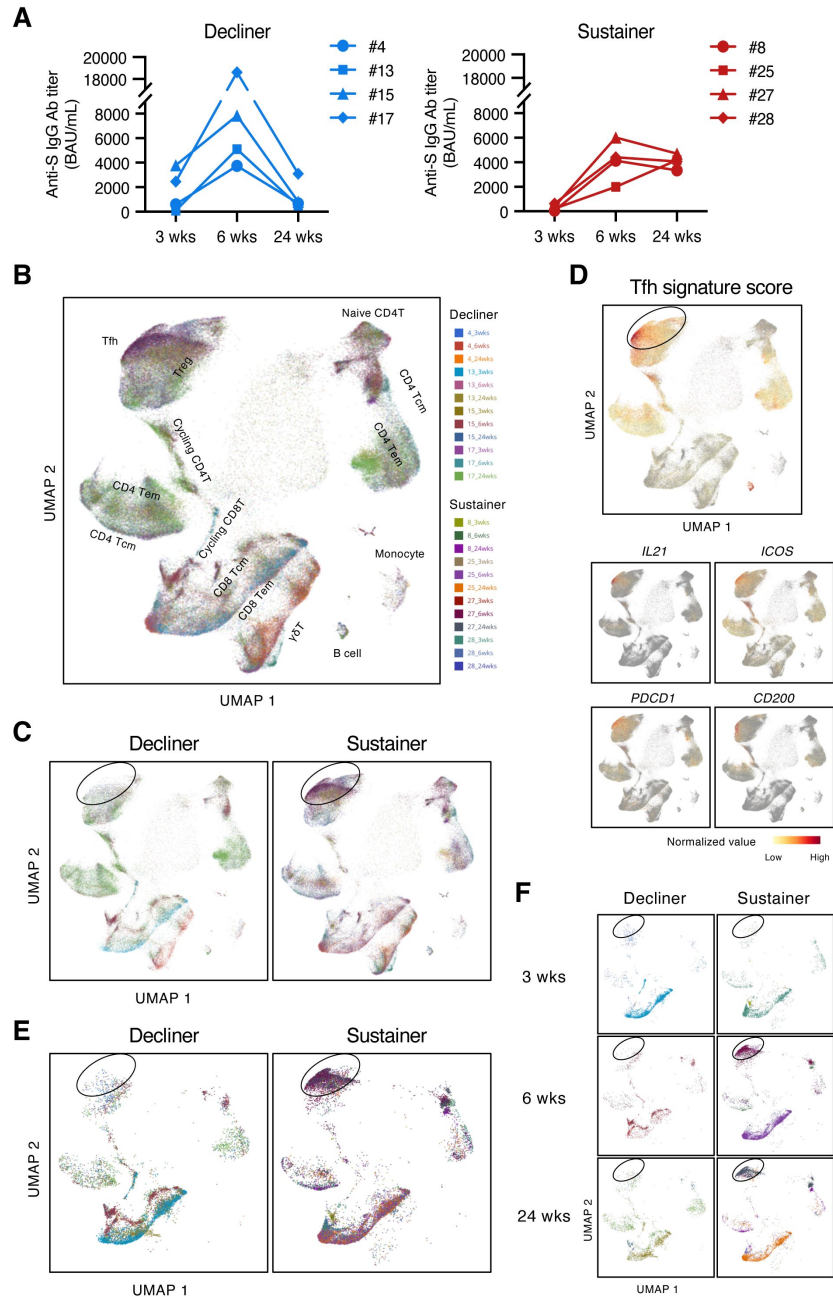


Fig. 2.

Antibody sustainers had highly expanded S-reactive Tfh clonotypes.

(A) Anti-S IgG titer of serum samples from sustainers and decliners is shown individually. (B, C, E, and F) UMAP projection of T cells in single-cell analysis of post-vaccinated samples collected from all donors. Each dot corresponds to a single cell and is colored according to the samples from different time points of donors. All samples together with annotated cell types (B), samples grouped by donor type (decliners and sustainers) (C), top 16 expanded clonotypes (16 clonotypes that had the most cell numbers from each donor) grouped by donor type (E), and top 16 expanded clonotypes grouped by time point and donor type (F) are shown. Tcm, central memory T cells; Tem, effector memory T cells; Treg, regulatory T cells; $\gamma\delta$ T, $\gamma\delta$ T cells. (D) Tfh signature score and expression levels of the canonical Tfh cell markers, *IL21*, *ICOS*, *PDCD1* and *CD200*, are shown as heat maps in the UMAP plot.

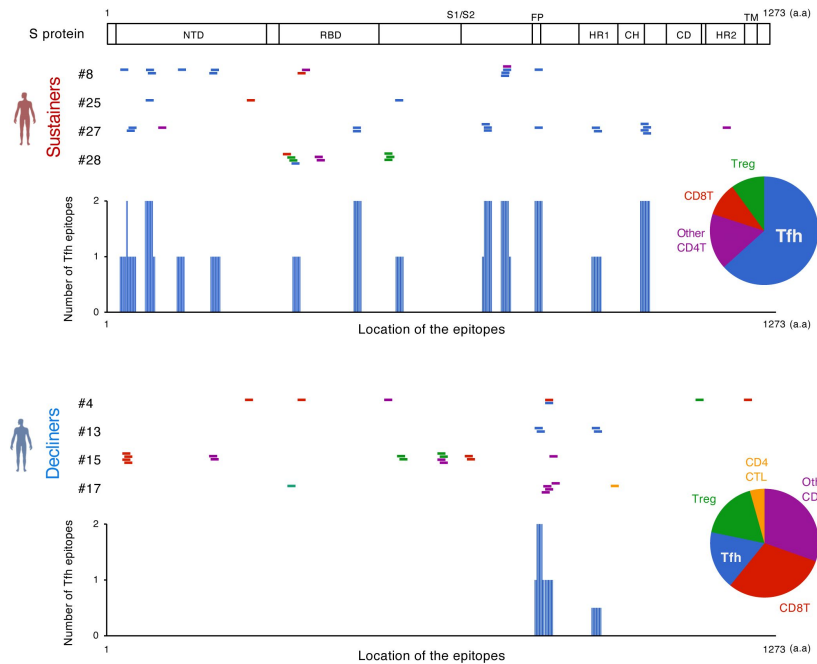


Fig. 3.

The location of S epitopes recognized by top expanded T clonotypes from post-vaccination samples.

T cell S epitopes recognized by top expanded TCR clonotypes in post-vaccinated samples from sustainers and decliners are mapped by their locations in S protein. Each short bar indicates a 15-mer peptide that activated the TCRs. Epitopes are shown in different colors according to the subsets of the T cells they activated. Relative frequencies of the T cell subsets are shown in pie charts. Numbers of identified epitopes recognized by a dominant T subset in sustainers (Tfh) are shown in blue bars. NTD, N-terminal domain; RBD, receptor-binding domain; FP, fusion peptide; HR1, heptad repeat 1; CH, central helix; CD, connector domain; HR2, heptad repeat 2; TM, transmembrane domain.

Donor	Clonotype	TRBV	CDR3β	TRBJ	TRAV	CDR3α	TRAJ	S epitope ^a	Restricting HLA
#8	Post_4	11-2	CASSPTGTNEKLFF	1-4	13-1	CAGGADGLTF	45	SFSTFKCYGVSPFKL ^{573-582b}	DRA-DRB1*15:02
	Post_5	19	CASSGRPEGPQHF	1-5	20	CAVLNQAGTALIF	15	FKIYSKHTPIN ²⁰¹⁻²¹¹	DRA-DRB1*09:01
	Post_6	11-2	CASSLEGTEAFF	1-1	5	CAESRYMGRRLTF	5	PQFCNDPFLGVYHYK ¹³³⁻¹⁴⁷	DPA1*01:03-DPB1*04:02
	Post_7	2	CAGLAGVDTGELFF	2-2	5	CAERVGRRLTF	5	YSVLYNSASFSTFKC ³⁶³⁻³⁷⁹	A*24:02
	Post_8	20-1	CSATRRRSYNEQFF	2-1	12-2	CAVLTNTGNQFYF	49	LLQYGSFCTQLNRL ⁷⁵³⁻⁷⁶⁷	DRA-DRB1*15:02
	Post_9	7-9	CASSLLGEQYF	2-7	22	CAGAGTSYGKLTFF	52	KRFDNPVLPFN ⁷⁷⁻⁸⁷	DPA1*02:02-DPB1*05:01
	Post_10	6-1	CASSEGASHPQHF	1-5	12-1	CVVNGSSASKIIF	3	LLQYGSFCTQL ⁷⁵³⁻⁷⁶³	DRA-DRB1*15:02
	Post_12	20-1	CSAYSINYEQFF	2-1	9-2	CALSMNTGFQKLVF	8	PPAYTNSPTRGVYYP ²⁵⁻³⁹	DRA-DRB1*09:01
	Post_14	19	CASRPNRGDNSPLHF	1-6	12-1	CVVSIIGFNVLHC	35	CSNLLQYGSFCTQL ⁷⁴⁹⁻⁷⁶³	DRA-DRB1*15:02
	Post_15	28	CASSLMGGAYGYTF	1-2	8-6	CAVRGGSGGSNYKLTF	53	SKRSFIEDLLFNKVT ⁸¹³⁻⁸²⁷	DPA1*01:03-DPB1*04:02
#25	Post_7	7-9	CAPSNANTGELFF	2-2	12-1	CVVNEADKLIF	34	YLQPTFLLLK ²⁶⁹⁻²⁷⁸	A*02:01
	Post_12	20-1	CSARDVEVGSGYTF	1-2	4	CLVGPYNQGGKLIF	23	TGVLTESNKKFLPFQ ⁵⁴⁹⁻⁵⁶³	DRA-DRB1*14:54
	Post_15	3-1	CASSPLSGSSYEQYF	2-7	12-1	CVVGTDSWGKLQF	24	TNGTKRFDNPVLPFN ⁷³⁻⁸⁷	DPA1*02:02-DPB1*05:01/ DPA1*01:03-DPB1*05:01
#27	Post_1	20-1	CSAIGADDTQYF	2-3	9-2	CALTSAGNKLTF	17	NQFNSAIGKIQ ⁹²⁵⁻⁹³⁵	DRA-DRB1*09:01
	Post_2	30	CAMNLGGHNPQHF	1-5	8-2	CVVSEASSYKLIF	12	SKRSFIEDLLFNKVT ⁸¹³⁻⁸²⁷	DPA1*02:02-DPB1*04:02
	Post_3	5-4	CASSQGQSGYGYTF	1-2	4	CLVGDSDTGRRLTF	5	NFTISVTTEIL ⁷¹⁷⁻⁷²⁷	DRA-DRB1*09:01
	Post_5	7-2	CASGTGSYNEQFF	2-1	12-2	CAVKRGNQGGKLIF	23	STEIYQAGSTPCNGV ⁴⁶⁹⁻⁴⁸³	DRA-DRB1*04:03
	Post_7	6-6	CASRLPGNRAQPHF	1-5	36/DV7	CAVESGSSNTGKLIF	37	KSNIIRGNIFGTTL ⁹⁷⁻¹¹¹	DRA-DRB4*01:03
	Post_8	6-5	CASSYSGGTGTGELFF	2-2	41	CAVGIRGNEKLTF	48	KVFRSSVLHST ⁴¹⁻⁵¹	DRA-DRB1*04:03
	Post_9	20-1	CSARDGQTATNEKLFF	1-4	17	CATNAGGTSYGKLTFF	52	EIRASANLAAT ¹⁰¹⁷⁻¹⁰²⁷	DRA-DRB1*04:03
	Post_11	30	CAMSVKGFPSQHF	1-5	6	CALGSTSNTGKLIF	37	EIRASANLAAT ¹⁰¹⁷⁻¹⁰²⁷	DRA-DRB1*04:03
	Post_13	5-6	CASSRTGYNSPLHF	1-6	27	CAGAGSGTYKYIF	40	STEIYQAGSTPCNGV ⁴⁶⁹⁻⁴⁸³	DRA-DRB1*04:03
	Post_15	5-5	CASSSDRNYGYTF	1-2	12-1	CVVNMVTGGYNKLIF	4	NFTISVTTEILFVSM ⁷¹⁷⁻⁷³¹	DRA-DRB1*09:01
#28	Post_16	7-9	CASSSQPLAGVKIGNEQFF	2-1	5	CAEIPPPSNTGKLIF	37	ISGINASVVNIQKEI ¹⁰⁶⁹⁻¹¹⁸³	DRA-DRB1*04:03
	Post_5	3-1	CASSQGGSEKLFF	1-4	1-1	CAVGGNTDKLIF	34	LVKNKCVNFNF ⁵³³⁻⁵⁴³	DRA-DRB3*03:01
	Post_10	12-3	CASSSGRTGFGYTF	1-2	30	CGTEPGSEKLVP	57	VIRGDEVQRQIA ⁴⁰¹⁻⁴¹¹	DRA-DRB3*03:01
	Post_11	5-8	CASSLQKTTGFSYGYTF	1-2	8-6	CAVSFYTGRRLTF	5	SVYAWNRKRIS ³⁴⁹⁻³⁵⁹	DRA-DRB1*13:02
	Post_12	18	CASSASVDPTEAFF	1-1	1-1	CASFPTGGGNKLTF	10	KSTNLVKMKCVNFNF ⁵²⁹⁻⁵⁴³	DRA-DRB3*03:01
	Post_14	7-6	CASSLSGTGGTGELEFF	2-2	4	CLVGDMRSGGGADGLTF	45	PFGEVFNATRFASVY ³³⁷⁻³⁵¹	B*40:01
	Post_15	6-2	CASSYPFSGGRGTGFEAFF	1-1	14/DV4	CAMRDIGFGNVLHC	35	WNRKRISMCVADYSV ⁵⁵³⁻⁵⁶⁷	DRA-DRB4*01:03

^aOverlapped epitope sequence is shown when a clonotype recognized two or three sequential peptides.

^bNumber ranges indicate the location of peptides in the proteins.

Table 4.

TCR clonotypes expanded in post-vaccinated samples and their TCR usages, epitopes and restricting HLAs.

Donor	Clonotype	TRBV	CDR3β	TRBJ	TRAV	CDR3α	TRAJ	S epitope	Restricting HLA
#4	Post_2	25-1	CASTGDNYGYTF	1-2	21	CAINTGNQFYF	49	YYVGYLQPR ²⁶⁵⁻²⁷³	A*33:03
	Post_10	7-9	CASRPSTGSRQYF	2-7	29	CAGNNAGNMLTF	39	FIKQYGDCLGDIAR ⁸³³⁻⁸⁴⁷	A*33:03
	Post_11	7-9	CASSTRTSGGGLSYEQYF	2-7	3	CAVNKAAGNKLF	17	YSVLYNSASFSTFKC ³⁶⁵⁻³⁷⁹	A*24:02
	Post_13	20-1	CSASIEQDGLGYTF	1-2	23/DV6	CAASIPNSGYALNF	41	FIKQYGDCLGDIAR ⁸³³⁻⁸⁴⁷	DQA1*01:02-DQB1*05:03
	Post_14	5-6	CASSPGQGILEQYF	2-7	24	CAFVPLSDGQKLLF	16	YIKWPWYIWL ¹²⁰⁹⁻¹²¹⁸	A*24:02
	Post_15	7-3	CASGIHTGELFF	2-2	26-1	CIVNNAGNMLTF	39	TDNTFVSGNCDVIG ¹¹⁷⁻¹³¹	DQA1*01:02-DQB1*06:04
#13	Post_16	7-6	CASSPGSEADTQYF	2-3	1-1	CAVRDGDGDKIIF	30	KSTNLVKNKCVNPF ⁵²⁸⁻⁵⁴³	DRA-DRB3*03:01
	Post_13	7-2	CASSVGSGKSGSAETQYF	2-5	22	CAVNEYSAGSYQLTF	28	SKRSFIEDLLFNKVT ¹³³⁻⁸²⁷	DPA1*01:03-DPB1*02:01
	Post_15	20-1	CSAGDTASTYGYTF	1-2	9-2	CALSDGAGNKLF	17	NQFNSAIGKIQ ⁹²³⁻⁹³⁵	DRA-DRB1*09:01
#15	Post_16	30	CAWSLQGRFPQH	1-5	38-1	CAPMKQGGSEKLVF	57	FIEDLLFNKVTLAD ⁸¹⁷⁻⁸³¹	DPA1*01:03-DPB1*02:01
	Post_1	12-4	CASSSHRDVGVEAFF	1-1	12-1	CVVNFDRSGTLGRLYF	18	TRGVYYPDKVF ⁵³⁻⁴³	B*15:01
	Post_6	3-1	CASSQGLNTGELFF	2-2	38-2/DV8	CAYRKTSGTYKYIF	40	WRVYSTGSNVF ⁶³³⁻⁶⁴³	DRA-DRB1*15:02
#17	Post_7	28	CASSFPDRYNSQPOHF	1-5	1-2	CAVRAVGGNKLVF	47	TRGVYYPDKVF ⁵³⁻⁴³	B*15:01
	Post_9	27	CASSPGHEQYF	2-7	14/DV4	CAMSPRTYKYIF	40	RSVASQSIIAY ⁶⁸⁵⁻⁶⁹⁵	B*15:01
	Post_11	3-1	CASSRELISEQYF	2-7	38-2/DV8	CAYRKTSGTYKYIF	40	WRVYSTGSNVF ⁶³³⁻⁶⁴³	DRA-DRB1*15:02
	Post_12	28	CASSSYGTSGGRAEQFF	2-1	16	CALSGGLTGGGNKLF	10	LGDIAARDLICAQKF ⁸⁴¹⁻⁸⁵⁵	DRA-DRB1*08:02
	Post_13	30	CAWRTGGGITSPLHF	1-6	8-2	CVVNNAGNMLTF	39	VFNIDGYFKIYSKH ¹⁹³⁻²⁰⁷	DPA1*02:02-DPB1*05:01
	Post_14	6-1	CASSEAGSGANVLTF	2-6	9-2	CALSGTGTYYKYIF	40	KKFLPFQQGR ⁵⁵⁷⁻⁵⁶⁷	DPA1*02:02-DPB1*05:01
	Post_16	27	CASSLGTINTGELFF	2-2	17	CATAPAGGTSGKLVF	52	IDGYFKIYSKHTPI ¹⁹⁷⁻²¹¹	DRA-DRB1*08:02
	Post_4	6-2	CASTTARGSYNEQFF	2-1	27	CAGHSNTGNQFYF	49	TRFASVYAWNRRKRIS ³⁴⁵⁻³⁵⁹	DRA-DRB1*08:02
	Post_10	9	CASSKTSYGAYNEQFF	2-1	9-2	CALDNARLMF	31	FIKQYGD ⁸³³⁻⁸³⁹	DRA-DRB1*15:01
	Post_11	20-1	CSARPFGGNGNEQFF	2-1	26-2	CILRDGTGANNLFF	36	QALNTLVKQLSSNF ^{G957-971}	DRA-DRB1*08:02
#17	Post_15	7-9	CASSLARGNSPLHF	1-6	38-2/DV8	CAFVGSQGNLIF	42	AARDLICAQKF ^{NGLT845-859}	DRA-DRB1*08:02

Table 4. (continued)

Donor	Clonotype	Mutated epitopes in VOCs	Domain	Response
#8	Post_4	Omicron BA.1 Omicron BA.2, 4/5	PFFTFKCYGVSPTKL ^a PFFAFKCYGVSPTKL	RBD ↓
		Omicron BA.1	FKIYSKHTPII	non-RBD ↑
#8	Post_6	Delta, Omicron BA.2, 4/5 Omicron BA.1	FQFCNDPFLDVYHK FQFCNDPFLD---HK	non-RBD ↓ ↓
		Omicron BA.1 Omicron BA.2, 4/5	YSVLYNLAPFFTFKC YSVLYNFAPFFAFKC	RBD ↓ ↓
#8	Post_8	Omicron BA.1, 2, 4/5	LLQYGSFCTQLKRAL	non-RBD ↑
#8	Post_10	Omicron BA.1, 2, 4/5	LLQYGSFCTQLKRAL	non-RBD ↑
#27	Post_5	Delta Omicron BA.1, 2, 4/5	STEIYQAGSKPCNGV STEIYQAGNKPCNGV	RBD ↓ ↓
		Delta Omicron BA.1, 2, 4/5	STEIYQAGSKPCNGV STEIYQAGNKPCNGV	RBD ↓ ↓
#28	Post_5	Omicron BA.1	LVKNKCVNFNGLK	non-RBD ↑
#28	Post_10	Omicron BA.2, 4/5	VIRGNEVSQIA	RBD ↓
#28	Post_14	Omicron BA.1, 2, 4/5	PFDEVFNATRFASVY	RBD ↓
#4	Post_11	Omicron BA.1 Omicron BA.2, 4/5	YSVLYNLAPFFTFKC YSVLYNFAPFFAFKC	RBD ↓ ↓
		Delta Omicron BA.1, 2, 4/5	RRRARSVASQSIIAY HRRARSVASQSIIAY	non-RBD ↑ ↑
#15	Post_16	Omicron BA.1	IDGYFKIYSKHTPII	non-RBD →
#17	Post_11	Omicron BA.1, 2, 4/5	QALNTLVKQLSSKFG	non-RBD ↓
#17	Post_15	Omicron BA.1	AARDLICAQKF ^a GLT	non-RBD ↓

^aAmino acids colored red indicate mismatches compared with corresponding S epitopes of Wuhan strain.

Table 5.

Reactivity of each clonotype to mutated epitopes in SARS-CoV-2 VOCs.

Identification of S epitopes and cross-reactive antigens of pre-existing T cell clonotypes

Before the pandemic, T cells cross-reacting to S antigen were present in the peripheral blood [13–17]. To characterize these pre-existing S-reactive cells, we analyzed the PBMCs collected from donors who consented to blood sample donation before vaccination (#4, #8, #13, #15, and #17). PBMCs were stimulated with the S peptide pool for 10 days, and proliferated T cells were sorted and analyzed by scTCR/RNA-seq. Similar to vaccine-induced S-reactive T cells (Fig. 2B), characteristics of pre-existing S-reactive T cells were diverse (Fig. 4A). To track the dynamics of cross-reactive clones after vaccination, we combined the single-cell sequencing data of pre- and post-vaccinated PBMCs and analyzed the clonotypes that have more than 50 cells in total (Fig. 4B). We did find some cross-reactive clonotypes that were further expanded by vaccination, and most of these clonotypes had cytotoxic features, being CD8⁺ effector memory T cells (Tem) or minor CD4⁺ cytotoxic T cells (CTLs). In contrast, most of the cross-reactive CD4⁺ T cells became minor clonotypes after vaccination.

We also explored the epitopes of the top 16 expanded clonotypes in each pre-vaccinated donor by reconstituting the TCRs into reporter cell lines. We identified 18 epitopes from S protein (Fig. 5 and Table 6) and determined some possible cross-reactive antigens. Most of these cross-reactive antigens originated from environmental or symbiotic microbes (Table 6). Furthermore, majority of the reactive T clonotypes showed regulatory T cell (Treg) signatures (Fig. 5). Six of these 80 analyzed clonotypes could also be frequently detected in the public TCR database Adaptive [23,24]. Notably, most of these clonotypes, except for one case, showed comparable frequencies between pre-pandemic healthy donors and COVID-19 convalescent patients (Fig. 6), suggesting that these clonotypes did not expand upon SARS-CoV-2 infection, despite they were present before the pandemic. Thus, it is unlikely that these cross-reactive T clonotypes contribute to the establishment of S-reactive T cell pools during either vaccination or infection.

Discussion

Previous studies showed that Tfh function and germinal center development were impaired in deceased COVID-19 patients [25] and Tfh cell number correlated with neutralizing antibody [26–28]. Consistent with the above studies, we found that the donors having sustained antibody titers between 6 to 24 weeks post-vaccination had more S antigen-responsive Tfh clonotypes maintained in the periphery as a memory pool. As circulating Tfh clonotypes can reflect the population of germinal center Tfh cells [29], it is possible that these maintained S-responsive Tfh cells contribute to the prolonged production of anti-S antibodies. These results imply that Tfh polarization of S-reactive T cells in the blood after 2nd vaccination can be a marker for the longevity of serum anti-S antibodies. Although monitoring of S-specific Tfh cells in germinal center is ideal [30], it is currently difficult for outpatients in clinics.

Since the antigen used for BNT162b2 is a full-length S protein from the Wuhan-Hu-1 strain, it is important to estimate whether vaccine-induced Wuhan S-reactive T cells recognize neutralizing antibody-evading VOCs, such as Omicron variants. Consistent with previous reports [31–33], most of the epitopes determined in the current study were conserved in Delta and Omicron (BA.1, BA.2 and BA.4/5) strains, suggesting that vaccine-induced T cells are able to recognize the mutated S proteins from these variants, despite B epitopes being largely mutated in these VOCs [31,32].

SARS-CoV-2-recognizing T cells existed prior to exposure of the S antigens [13–17], which is consistent with our observation with PBMCs from donors who were uninfected and pre-vaccinated. Among these pre-existing S-reactive clonotypes, CD8⁺ cytotoxic T clonotypes were expanded by the vaccination, whereas most of CD4⁺ T clonotypes became less dominant after

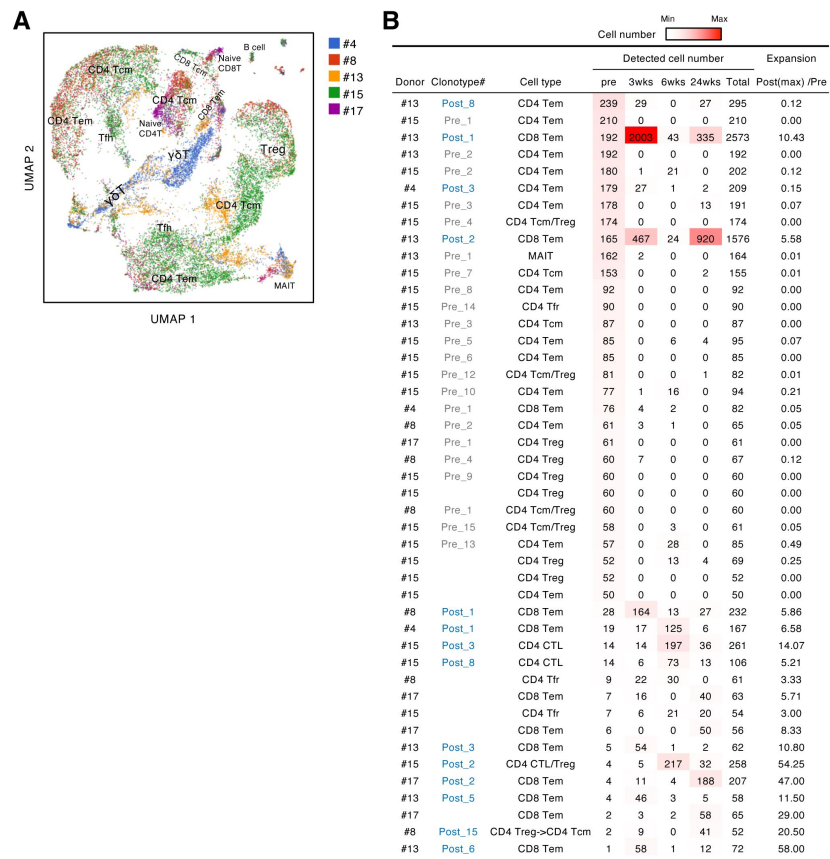


Fig. 4.

Characteristics and dynamics of S-cross-reactive clonotypes.

(A) UMAP projection of T cells in single-cell analysis of pre-vaccinated T cells from donors #4, #13, #15, #17, and #8. Each dot corresponds to a single cell and is colored according to the samples from different donors. Annotated cell types are shown.

(B) Donor, name of reconstituted clonotypes, cell type, number in different time points, and expansion ratio of clonotypes that were found in pre-vaccinated samples and had more than 50 cells in the combined pre- and post-vaccinated sample set. For clonotypes that showed more than one type, the major type is listed in the front. The expansion ratio was calculated using the maximum cell number at post-vaccination points divided by the cell number at the pre-vaccination point of each clonotype. Clonotypes that have an expansion ratio larger than 1 are considered as expanded post-vaccination. Cell numbers at individual time points are shown as heat map. Tfr, follicular regulatory T cells; MAIT, mucosal-associated invariant T cells.

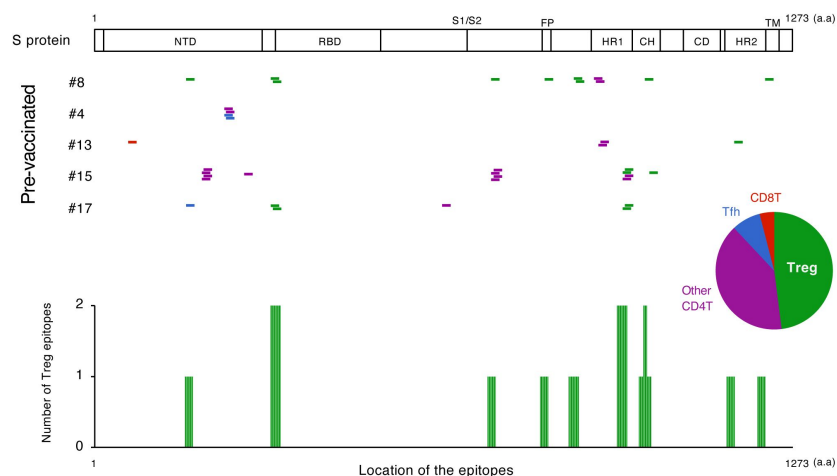


Fig. 5.

The location of S epitopes of pre-existing S-reactive T cells.

S epitopes recognized by top expanded TCR clonotypes in pre-vaccinated samples are mapped by their locations in S protein. Each short bar indicates a 15-mer peptide that activated the TCRs. Epitopes are shown in different colors according to the subtypes of the T cells they activated. Relative frequencies of the T cell subtypes from all five donors are shown in the pie chart. Numbers of identified epitopes recognized by a dominant T subset of pre-existing clonotypes (Treg) from all donors are shown in green bars.

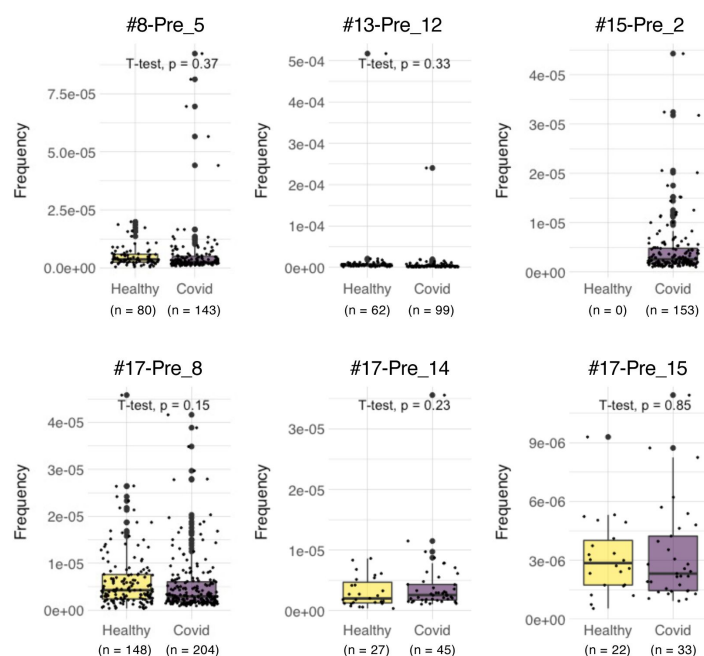


Fig. 6.

Frequencies of pre-existing S-reactive clonotypes in the public database of uninfected and infected cohorts.

TCR β sequences of the top expanded clonotypes in pre-vaccinated samples were investigated in the Adaptive database. Frequencies of detected clonotypes are shown in box plot. Healthy, dataset from 786 healthy donors. COVID, dataset from 1487 COVID-19 patients.

Donor	Clonotype	TRBV	CDR3p	TRBJ	TRAV	CDR3a	TRAJ	S epitope	Restricting HLA	Cross-reactive antigen [species]	Cross-reactive peptide	Post-vaccinated expansion
#4	Pre_5	6-6	CASSYPGGGSETQYF	2-5	35	CAGVAVQGAQLVF	54	LIALHRSYLTP ₂₀₋₂₅ ^a	DRA-DRB1*14:54	Phosphoribosylformylglycinamide cyclo-ligase [<i>Firmicutes bacterium</i>]	VAEALLAVHRSYLTP ₂₀₋₂₅ ^b	No
#4	Pre_7	6-6	CASSYPGGSGGELFF	2-2	21	CAVENSGNTPLVF	29	LIALHRSYLTP ₂₀₋₂₅	DQA1*01:04-DQB1*05:03	Phosphoribosylformylglycinamide cyclo-ligase [<i>Firmicutes bacterium</i>]	VAEALLAVHRSYLTP ₂₀₋₂₅	No
#8	Pre_1	6-2	CASRPNRGFRGNQPQHF	1-5	23/IV6	CAGEEKEKSGSRLTF	58	NCTFEVVSQPLMDL ₁₀₅₋₁₇₉	DRA-DRB1*15:01	Fumarylacetoacetate hydrolase family protein [<i>Akkalkomonas facialis</i>] Hypothetical protein [<i>Planomicrobium bacterium</i>]	ASLIEVVSQPFLLP ₂₅₋₂₉ AAGFEVVSQPFSLP ₁₀₃₋₁₀₇	No
#8	Pre_2	6-1	CASIRDRVADTQYF	2-3	30	CGTETDWSGKLQF	24	RFNGIGVTQNV ₄₀₅₋₄₁₅	DQA1*03:02-DQB1*03:03	SEL-1-like repeat protein [<i>Bacteroides bacterium</i>] ^c	LGVTYFNGIGVTQDQ ₂₆₋₃₀	No
#8	Pre_3	27	CATKGEANYGYTF	1-2	12-3	CAMSEMGTFQKLVF	8	SIVRFPNITNL ₁₀₃₋₁₀₅	DRA-DRB1*15:01	LTA synthase family protein [<i>Dactylosporangium dentifilum</i>]	LPKGSVVRNPNITNL ₁₀₃₋₁₀₄	No
#8	Pre_5	5-1	CASSLRTGELFF	2-2	8-1	CAVNGRNTGFQKLVF	8	NFTISVTTEILPVSM ₁₁₃₋₁₂₁	DRA-DRB1*09:01	Major capsid protein [<i>Human papillomavirus 143</i>] Periplasmic trehalase [<i>Chlamydia bacterium</i>]	NFTISVTTDAGDINE ₁₀₆₋₁₀₄ LSTIVTTEILPVDL ₁₀₈₋₁₀₁	No
#8	Pre_9	7-2	CASAAGTGTGETQYF	2-5	5	CAETPFLSGTYKYIF	40	YIKWPWYIWLGFIAQ ₁₀₈₋₁₁₂	DRA-DRB3*01:02	Spike glycoprotein [<i>Human coronavirus HKU1</i>]	VKHPWYVVLIIISFSP ₁₀₁₋₁₀₁	No
#8	Pre_10	6-6	CASSLQGGIHEQYF	2-7	26-1	CIVERGGGSNYKLTFF	53	SKRSFIEDLLFNKVT ₁₀₃₋₁₀₇	DPA1*01:03-DPB1*04:02	Hypothetical protein, partial [<i>Acinetobacter baumannii</i>] Spike protein [<i>Feline coronavirus</i>] Spike protein [<i>Canine coronavirus</i>]	GKRSAVEDLLFNKVV ₁₀₄₋₁₁₄ GKRSAVEDLLFNKVV ₁₀₆₋₁₀₄ GKRSAVEDLLFNKVV ₁₀₇₋₁₀₁	No
#8	Pre_14	4-3	CASSQRQAGDQYF	2-3	19	CALSEAGIQGAQLVF	54	IDRLITGRQLSQTYY ₉₀₃₋₁₀₀₇	DQA1*01:03-DQB1*06:01	Exonuclease ABC subunit UvrA [<i>Leptospira bacterium</i>]	VDRLITGRLESSRLR ₁₀₈₋₁₁₀	No
#8	Pre_15	20-1	CSAKDRIYGYTF	1-2	26-1	CIVRSPSGSARQLTF	22	MIAQTTSALLA ₄₈₀₋₄₇₉	DRA-DRB1*15:01	MATE family efflux transporter [<i>Solenomonas nectis</i>]	ATIIAQYTSALLALR ₃₄₂₋₂₉₆	No
#13	Pre_5	4-3	CASSQVSTGTGITGANVLTF	2-6	5	CARRSSASKIIF	3	QNVLYENQKLIT ₁₀₃₋₁₀₃	DRA-DRB3*01:02	Hypothetical protein [<i>Neobacillus virus</i>]	TNVLYENQKLINLF ₁₀₄₋₁₀₃	No
#13	Pre_8	18	CASSPRAPPYEQYF	2-7	21	CAVRPAGGTGNQYF	49	DEYFKHSTSPVDVLG ₁₁₀₃₋₁₁₀₇	DRA-DRB1*15:01	Type VI secretion system contractile sheath large subunit [<i>Salmonella enterica</i>]	DYTFDHTSPVDVLG ₁₁₀₃₋₁₁₀₁	No
#13	Pre_12	4-2	CASSQEGNTEAFF	1-1	20	CGCRGGTSYKLTFF	52	NVTWFHAIHVSQGTNG ₄₁₋₇₅	A*02:07	Dihydrofolate synthase [<i>Actinobaculum sp. 313</i>]	PQRSFTHAIHVTGTNG ₄₁₋₄₅	No
#15	Pre_1	20-1	CSARDLTASAHGYTF	1-2	17	CATDAGQGGKLIFF	23	SVTTEILPVSM ₁₀₃₋₁₁₀	DQA1*01:03-DQB1*06:01	Hypothetical protein [<i>Mycobacterium bacterium</i>]	PVTTEILPVSDPPG ₁₀₃₋₁₀₉	No
#15	Pre_2	24-1	CATSDLPQPHF	1-5	16	CALSGVGSQYSTLTF	11	SVTTEILPVSM ₁₀₃₋₁₁₀	DQA1*01:03-DQB1*06:01	Hypothetical protein [<i>Mycobacterium bacterium</i>]	PVTTEILPVSDPPG ₁₀₃₋₁₀₉	No
#15	Pre_3	6-1	CASDPKNGGEQYF	2-7	29/IV5	CAASVGVGFNVLEH	35	FKIYSKHTPIIN ₁₀₆₋₁₂₁	DRA-DRB3*01:02	Uncharacterized protein APUU_312895 [<i>Aspergillus paulanensis</i>]	CRAAFKLYSKHTFPV ₁₀₅₋₁₁₇	No
#15	Pre_4	19	CASGLAGGNTGELFF	2-2	10	CVPSGGYGNKLIF	4	QALNTLVKQLS ₉₀₇₋₉₀₇	DRA-DRB1*08:02	4-hydroxybenzoate octaprenyltransferase [<i>Pseudomonas aeruginosa</i>]	IQPLNTLVKQLSVAA ₁₁₃₋₁₁₆	No
#15	Pre_5	6-5	CASSAGLAGGNTQYF	2-3	5	CAVIGSGARQLTF	22	QALNTLVKQLS ₉₀₇₋₉₀₇	DRA-DRB1*08:02	4-hydroxybenzoate octaprenyltransferase [<i>Pseudomonas aeruginosa</i>]	IQPLNTLVKQLSVAA ₁₁₃₋₁₁₆	No
#15	Pre_6	2	CASVQNGEQQF	2-1	9-2	CALTRFVGGATNKLIF	32	RTFLKYNENGTTIT ₂₇₃₋₂₈₇	DRA-DRB1*15:01	Unnamed protein product [<i>Mycobacterium</i>]	NKKLLKYNENGTTIT ₂₇₃₋₂₈₁	No
#15	Pre_7	4-1	CASSHDGTPPDQYF	2-3	29/IV5	CAAYSNYQLIW	33	FKIYSKHTPIIN ₁₀₆₋₁₂₁	DRA-DRB1*15:01	Uncharacterized protein APUU_312895 [<i>Aspergillus paulanensis</i>]	CRAAFKLYSKHTFPV ₁₀₅₋₁₁₇	No
#15	Pre_15	2	CASSETGRGTDQYF	2-3	9-2	CALYRGTYKYIF	40	LQSLQTYVTQQLIRA ₁₀₃₋₁₀₃	DRA-DRB1*15:01	Dyp-type peroxidase [<i>Acinetobacter sp.</i>]	CTVLQTYVTQQLSV ₁₀₄₋₁₀₄	No
#17	Pre_7	6-1	CASSLRGAFGYTF	1-2	35	CAGHLYGSGQGNLIF	42	NCTFEVVSQPLMDL ₁₀₅₋₁₇₉	DPA1*01:03-DPB1*04:02	Fumarylacetoacetate hydrolase family protein [<i>Akkalkomonas facialis</i>] Hypothetical protein [<i>Planomicrobium bacterium</i>]	ASLIEVVSQPFLLP ₂₅₋₂₉ AAGFEVVSQPFSLP ₁₀₃₋₁₀₇	No
#17	Pre_8	5-1	CASSLNSGANVLTF	2-6	13-1	CAASIVQDQKLVF	8	LTPTRVRYSTGSGNVF ₁₂₉₋₁₄₃	DRA-DRB1*08:02	Hypothetical protein [<i>Neorhizoglyphus chlorosacciniferae</i>]	APGTPTRVRYSTART ₁₂₉₋₁₃₁	No
#17	Pre_14	5-1	CASSLGLAGLYNEQFF	2-1	38-1	CAFINNAGNNMLTF	39	QALNTLVKQLS ₉₀₇₋₉₀₇	DRA-DRB1*08:02	4-hydroxybenzoate octaprenyltransferase [<i>Pseudomonas aeruginosa</i>]	IQPLNTLVKQLSVAA ₁₁₃₋₁₁₆	No
#17	Pre_15	7-2	CASSRTSGGTYEQYF	2-7	25	CAGQNTDKLIFF	34	SIVRFPNITNL ₁₀₃₋₁₀₅	DRA-DRB1*15:01	LTA synthase family protein [<i>Dactylosporangium dentifilum</i>]	LPKGSVVRNPNITNL ₁₀₃₋₁₀₄	Yes

^aNumber ranges indicate the location of peptides in the proteins.

^bAmino acids colored red indicate mismatches compared with corresponding S epitopes of Wuhan strain.

^cAntigen names and peptide sequences in light gray indicate inactive antigens of the corresponding T clonotypes.

Table 6.

S-cross-reactive TCR clonotypes expanded in pre-vaccinated samples and their TCR usages, epitopes, restricting HLAs and cross-reactive epitopes in microbes other than SARS-CoV-2.

vaccination (**Fig. 4B**). Currently, the reason for the opposite tendency is unclear. In the present study, we showed that pre-existing T clonotypes cross-reacting to S protein are unlikely to contribute to vaccine-driven T cell immunity. This could be due to the fact that cross-reactive T cells had relatively low avidity to S protein [34]. Alternatively, but not mutually exclusively, considering that most of these cross-reactive T clonotypes have Treg signature (**Fig. 5**), they could be developed to tolerate symbiotic or environmental antigens, and might be ineffective to the defense against SARS-CoV-2 and thus replaced by the other effective T clonotypes induced by vaccination. One exceptional pre-existing clonotype was #15-Pre_2, as they vigorously expanded in COVID-19 patients (**Fig. 6**). This clonotype was clustered within a CD4⁺ Tem population and cross-reactive to environmental bacteria, *Myxococcales bacterium* (**Table 6**). Thus, in some particular settings, clonotypes primed by common bacterial antigens might potentially contribute during infection.

Common cold human coronavirus (HCoV)-derived S proteins are reported as potential cross-reactive antigens for pre-existing SARS-CoV-2 S-reactive T cells [15,18–20]. However, the highly responding SARS-CoV-2 S-reactive clonotypes in pre-vaccinated donors rarely reacted with HCoV S proteins in the present study, which might be partly due to the difference of cohorts or ethnicities. Instead, most of those T cells cross-reacted with environmental or symbiotic bacteria. These observations suggest that these cross-reactive T cells might have been developed to establish tolerance against less harmful microbes, and thus unlikely to efficiently contribute to the protective viral immunity. Vaccination may induce opposite tendencies on T cell clonotypes that recognize the same antigen [35], which is hardly detected by the bulk T cell analyses. The current study highlights the necessity of dynamic tracing of T cell responses in an epitope-specific clonotype resolution for the evaluation of vaccine-induced immunity.

This study suggests that mRNA vaccine is potent enough to prime rare T cell clonotypes that become dominant afterwards. Furthermore, we propose that the types of CD4⁺ T clonotypes developed shortly after two doses of vaccination could be an indication of the longevity of antibodies. Tfh-inducing adjuvants or Tfh-skewing epitope would be a promising “directional” booster in the post-vaccine era when most people worldwide were exposed with the same antigen in multiple doses within a short period. Furthermore, in addition to SARS-CoV-2, this strategy can also be applicable for the prevention of other infectious diseases of which neutralizing antibody titers are effective for protection.

Materials and Methods

Ethics statement and sample collection

This project was approved by Osaka University Institutional Review Board (IRB) (reference No. 21487). 43 volunteers were enrolled in this project. Informed consent was obtained from all participants before the first blood sampling. Samples (serum, whole blood, and PBMCs) were collected four times at 0–7 days before 1st dose vaccination as pre-vaccination, at 14–21 days after 1st dose vaccination as 3 weeks sample, at 35–49 days after 1st dose vaccination as 6 weeks sample, and at 154–182 days after 1st dose of vaccination as 24 weeks sample. At the same time of blood sampling, adverse event information was also collected from all participants. PBMCs were isolated using BD vacutainer® CPT™ cell separation tube (Beckton Dickinson), according to manufacturers’ instructions. Isolated PBMCs were stored in the vapor phase of liquid nitrogen until use.

Antibody titer determination by enzyme-linked immunosorbent assay (ELISA)

Serum antibody titer was measured using ELISA. Briefly, recombinant ancestral S protein (S1+S2, Cell Signaling Technology; 1 µg/ml) or recombinant nucleocapsid protein (Acrobiosystems; 1 µg/ml) was coated on 96-well plate at 4 °C overnight. On the second day, wells were blocked with blocking buffer (PBS-T (0.05% tween@20) containing 5% skim milk) for 2 h at room temperature. The sera were diluted from 10 to 31,250 folds in blocking buffer and incubated overnight at 4 °C. The next day, wells were washed and incubated with horseradish peroxidase (HRP)-conjugated antibodies (GE Healthcare) for 3 h at room temperature. After being washed with PBS-T, wells were incubated with the peroxidase chromogenic substrate 3,3'-5,5'-tetramethyl benzidine (Sigma-Aldrich) for 30 min at room temperature, then the reaction was stopped by 0.5 N sulfuric acid (Sigma Aldrich). The absorbance of wells was immediately measured at 450 nm with a microplate reader (Bio-Rad).

The value of the half-maximal antibody titer of each sample was calculated from the highest absorbance in the dilution range by using Prism 8 software. The calculated antibody titer was converted to BAU/ml by using WHO International Standard 20/136 (NIBSC) for ancestral S-specific antibody titer.

Whole blood interferon-gamma release immune assay (IGRA) for SARS-CoV-2 specific T cell responses using QuantiFERON

SARS-CoV-2 specific T cell immune responses were evaluated by QuantiFERON SARS-CoV-2 (Qiagen) [36], according to manufacturer's instructions, in which CD4⁺ T cells were activated by epitopes coated on Ag1 tube, and CD4⁺ and CD8⁺ T cells were activated by epitopes coated on Ag2 tube. Briefly, 1 ml of whole blood sample with heparin is added into each of Nil (negative control), Mito (positive control), Ag1, and Ag2 tubes, and incubated at 37 °C for 22–24 h. Tubes were then centrifuged at 3,000 × g for 15 min for collecting plasma samples. IFN γ derived from activated T cells was measured with enzyme-linked immunosorbent assay (ELISA) (Qiagen) according to the manufacturer's instructions. IFN γ concentration (IU/ml) was calculated with background (Nil tube) subtracted from values of Ag1 or Ag2 tubes.

Pseudo-typed virus neutralization assay

The neutralizing activity of serum antibodies was analyzed with pseudo-typed VSVs as previously described [37]. Briefly, Vero E6 cells stably expressing TMPRSS2 were seeded on 96-well plates and incubated at 37 °C for 24 h. Pseudoviruses were incubated with a series of dilutions of inactivated serum for 1 h at 37 °C, then added to Vero E6 cells. At 24 h after infection, cells were lysed with cell culture lysis reagent (Promega), and luciferase activity was measured by Centro XS³ LB 960 (Berthold).

In vitro stimulation of PBMCs

Cryopreserved PBMCs were thawed and washed with warm RPMI 1640 medium (Sigma) supplemented with 5% human AB serum (GeminiBio), Penicillin (Sigma), streptomycin (MP Biomedicals), and 2-mercaptoethanol (Nacalai Tesque). PBMCs were labeled with Cell Proliferation Kit (CellTrace™ Violet, ThermoFisher) following the manufacturer's protocol and were stimulated in the same medium with S peptide pool (1 µg/ml per peptide, IPT) for 10 days, with human recombinant IL-2 (1 ng/ml, Peprotech), IL-7 (5 ng/ml, BioLegend) and IL-15 (5 ng/ml, Peprotech) supplemented on day 2, day 5 and day 8 of the culture. On day 10 cells were washed and stained with anti-human CD3 and TotalSeq-C Hashtags antibodies. Proliferated T cells (CD3⁺CTV^{low}) were sorted by cell sorter SH800S (SONY) and used for single-cell TCR and RNA sequencing analyses.

Single cell-based transcriptome and TCR repertoire analysis

Single cell library was prepared using the reagents from 10x Genomics following the manufacturer's instructions. After reverse transcription, cDNA was amplified for 14 cycles, and up to 50 ng of cDNA was used for construction of gene expression and TCR libraries. Libraries were sequenced in paired-end mode, and the raw reads were processed by Cell Ranger 3.1.0 (10x Genomics). Doublets and empty drops were removed by using Scrublet [38] and gating out the events whose main hashtag reads are less than 95% of the total hashtag reads. The top 4000 highly variable genes were used for clustering. Tfh signature score was generated using canonical Tfh marker genes (*IL21*, *ICOS*, *CD200*, *PDCD1*, *POU2AF1*, *BTLA*, *CXCR5*, and *CXCL13*) and UMAP plots were exported using BBrowser [39].

Reporter cell establishment and stimulation

TCR α and β chain cDNA sequences were introduced into a mouse T cell hybridoma lacking TCR and having a nuclear factor of activated T-cells (NFAT)-green fluorescent protein (GFP) reporter gene [40] using retroviral vectors. TCR-reconstituted cells were co-cultured with 1 μ g/ml of peptides in the presence of antigen-presenting cells (APCs). After 20 h, cell activation was assessed by GFP and CD69 expression.

Antigen-presenting cells

Transformed B cells and HLA-transfected HEK293T cells used as APCs were generated as described [21]. For transformed B cells, 3×10^5 PBMCs were incubated with the recombinant Epstein-Barr virus (EBV) suspension [41] for 1 h at 37°C with mild shaking every 15 min. The infected cells were cultured in RPMI 1640 medium supplemented with 20% fetal bovine serum (FBS, CAPRICORN SCIENTIFIC GmbH) containing cyclosporine A (CsA, 0.1 μ g/ml, Cayman Chemical). Immortalized B lymphoblastoid cell lines were obtained after 3 weeks of culture and used as APCs. For HLA-transfected HEK293T cells, plasmids encoding HLA class I/II alleles [42] were transfected in HEK293T cells with PEI MAX (Polysciences).

Determination of epitopes and restricting HLA

15-mer peptides with 11 amino acids overlap that cover the full length of S protein of SARS-CoV-2 were synthesized (GenScript). Peptides were dissolved in DMSO at 12 mg/ml and 12-15 peptides were mixed to create 26 different semi-pools. TCR-reconstituted reporter cells were stimulated with 1 μ g/ml of S peptide pool (1 μ g/ml per peptide, JPT), then 36-peptide pools that consist of 3 semi-pools each, then semi-pools, and then 12 individual peptides in the presence of autologous B cells to identify epitope peptides. To determine the restricting HLA, HLAs were narrowed down by co-culturing reporter cells with autologous and various heterologous B cells in the presence of 1 μ g/ml of the epitope peptide. HLAs shared by activatable B cells were transduced in HEK239T cells and used for further co-culture to identify the restricting HLA.

Statistics

All values with error bars are presented as the mean $\pm$ SEM. One-way ANOVA followed by Turkey's post hoc multiple comparison test was used to assess significant differences in each experiment using Prism 8 software (GraphPad Software). Differences were considered to be significant when *P* value was less than 0.05. *P* values in Fig. 6 were calculated with t-test using the "stat_compare_means" function in R.

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Editors

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Reviewer #1 (Public Review):

- A summary of what the authors were trying to achieve.

The authors cultured pre- and Post-vaccine PBMCs with overlapping peptides encoding S protein in the presence of IL-2, IL-7, and IL-15 for 10 days, and extensively analyzed the T cells expanded during the culture; by including scRNAseq, scTCRseq, and examination of reporter cell lines expressing the dominant TCRs. They were able to identify 78 S epitopes with HLA restrictions (by itself represents a major achievement) together with their subset, based on their transcriptional profiling. By comparing T cell clonotypes between pre- and post-vaccination samples, they showed that a majority of pre-existing S-reactive CD4⁺ T cell clones did not expand by vaccinations. Thus, the authors concluded that highly-responding S-reactive T cells were established by vaccination from rare clonotypes.

- An account of the major strengths and weaknesses of the methods and results.

Strengths

- Selection of 4 "Ab sustainers" and 4 "Ab decliners" from 43 subjects who received two shots of mRNA vaccinations.
- Identification of S epitopes of T cells together with their transcriptional profiling. This allowed the authors to compare the dominant subsets between sustainers and decliners.

Weaknesses

- Fig. 3 provides the epitopes, and the type of T cells, yet the composition of subsets per subject was not provided. It is possible that only one subject out of 4 sustainers expressed many Tfh clonotypes and explained the majority of Tfh clonotypes in the sustainer group. To exclude this possibility, the data on the composition of the T cell subset per subject (all 8 subjects) should be provided.
- S-specific T cells were obtained after a 10-day culture with peptides in the presence of multiple cytokines. This strategy tends to increase a background unrelated to S protein. Another shortcoming of this strategy is the selection of only T cells amenable to cell proliferation. This strategy will miss anergic or less-responsive T cells and thus create a bias in the assessment of S-reactive T cell subsets. This limitation should be described in the Discussion.
- Fig. 5 shows the epitopes and the type of T cells present at baseline. Do they react to HCoV-derived peptides? I guess not, as it is not clearly described. If the authors have the data, it

should be provided.

- As the authors discussed (L172), pre-existing S-reactive T cells were of low affinity. The raw flow data, as shown in Fig. S3, for pre-existing T cells may help discuss this aspect.

Reviewer #2 (Public Review):

Summary: A short-term comparison of durability of S antibody levels after 2-dose vaccination, showing that better or more poorly sustained responses correlate with the presence of Tfh cells.

Strengths:

Novelty of approach in expanding, sequencing and expressing TCRs for functional studies from the implicated populations.

Weaknesses:

Somewhat outdated question, short timeline, small numbers, over-interpretation of sequence homology data.

Reviewer #3 (Public Review):

Summary: The paper aims to investigate the relationship between anti-S protein antibody titers with the phenotypes&clonotypes of S-protein-specific T cells, in people who receive SARS-CoV2 mRNA vaccines. To do this, the paper recruited a cohort of Covid-19 naive individuals who received the SARS-CoV2 mRNA vaccines and collected sera and PBMCs samples at different timepoints. Then they mainly generate three sets of data: 1). Anti-S protein antibody titers on all timepoints. 2) Single-cell RNAseq/TCRseq dataset for divided T cells after stimulation by S-protein for 10 days. 3) Corresponding epitopes for each expanded TCR clones. After analyzing these results, the paper reports two major findings & claims: A) Individuals having sustained anti-S protein antibody response also have more so-called Tfh cells in their single-cell dataset, which suggests Tfh-polarization of S-specific T cells can be a marker to predict the longevity of anti-S antibody. B). S-reactive T cells do exist before the vaccination, but they seem to be unable to respond to Covid-19 vaccination properly.

The paper's strength is it uses a very systemic and thorough strategy trying to dissect the relationship between antibody titers, T cell phenotypes, TCR clonotypes and corresponding epitopes, and indeed it reports several interesting findings about the relationship of Tfh/sustained antibody and about the S-reactive clones that exist before the vaccination. However, the main weakness is these interesting claims are not sufficiently supported by the evidence presented in this paper. I have the following major concerns:

1. The biggest claim of the paper, which is the acquisition of S-specific Tfh clonotypes is associated with the longevity of anti-S antibodies, should be based on proper statistical analysis rather than just a UMAP as in Fig2 C, E, F. The paper only shows the pooled result, but it looks like most of the so-called Tfh cells come from a single donor #27. If separating each of the 4 decliners and sustainers and presenting their Tfh% in total CD4+ T cells respectively, will it statistically have a significant difference between those decliners and sustainers? I want to emphasize that solid scientific conclusions need to be drawn based on proper sample size and statistical analysis.

2. The paper does not provide any information to justify its cell annotation as presented in Fig 2B, 4A. Moreover, in my opinion, it is strange to see that there are two clusters of cells sit on both the left and right side of UMAP in Fig2B but both are annotated as CD4 Tcm and Tem. Also Tfh and Treg belong to a same cluster in Fig 2B but they should have very distinct transcriptomes and should be separated nicely. Therefore I believe the paper can be more convincing if it can present more information and discussion about the basis for its cell annotation.
3. Line 103-104, the paper claims that the Tfh cluster likely comes from cTfh cells. However considering the cells have been cultured/stimulated for 10 days, cTfh cells might lose all Tfh features after such culture. To my best knowledge there is no literature to support the notion that cTfh cells after stimulated in vitro for 10 days (also in the presence of IL2, IL7 and IL15), can still retain a Tfh phenotype after 10 days. It is possible that what actually happens is, instead of having more S-specific cTfh cells before the cell culture, the sustainers' PBMC can create an environment that favors the Tfh cell differentiation (such as express more pro-Tfh cytokines/co-stimulations). Thus after 10-days culture, there are more Tfh-like cells detected in the sustainers. The paper may need to include more evidence to support cTfh cells can retain Tfh features after 10-days' culture.
4. It is in my opinion inaccurate to use cell number in Fig4B to determine whether such clone expands or not, given that the cell number can be affected by many factors like the input number, the stimulation quality and the PBMC sample quality. A more proper analysis should be considered by calculating the relative abundance of each TCR clone in total CD4 T cells in each timepoint.
5. It is well-appreciated to express each TCR in cell line and to determine the epitopes. However, the author needs to make very sure that this analysis is performed correctly because a large body of conclusions of the paper are based on such epitope analysis. However, I notice something strange (maybe I am wrong) but for example, Table 4 donor #8 clonotype post_6 and _7, these two clonotypes have exactly the same TRAV5 and TRAJ5 usage. Because alpha chain don't have a D region, in theory these clonotypes, if have the same VJ usage, they should have the same alpha chain CDR3 sequences, however, in the table they have very different CDR3 α aa sequences. I wish the author could double check their analysis and I apologize in advance if I raise such questions based on wrong knowledge.

Author Response

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- A summary of what the authors were trying to achieve.

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We thank the reviewer for this comment. We will show the data in the revised manuscript.

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We will describe the limitation and advantage of our strategy in the revised manuscript.

- *Fig. 5 shows the epitopes and the type of T cells present at baseline. Do they react to HCoV-derived peptides? I guess not, as it is not clearly described. If the authors have the data, it should be provided.*

We apologize for not mentioning it clearly. As we have confirmed the unresponsiveness using synthetic HCoV peptides, we will include these data in the revised manuscript.

- *As the authors discussed (L172), pre-existing S-reactive T cells were of low affinity. The raw flow data, as shown in Fig. S3, for pre-existing T cells may help discuss this aspect.*

We thank the reviewer for this helpful comment. We will add the discussion to the revised manuscript.

Reviewer #3 (Public Review):

Summary: The paper aims to investigate the relationship between anti-S protein antibody titers with the phenotypes&clonotypes of S-protein-specific T cells, in people who receive SARS-CoV2 mRNA vaccines. To do this, the paper recruited a cohort of Covid-19 naive individuals who received the SARS-CoV2 mRNA vaccines and collected sera and PBMCs samples at different timepoints. Then they mainly generate three sets of data: 1). Anti-S protein antibody titers on all timepoints. 2) Single-cell RNAseq/TCRseq dataset for divided T cells after stimulation by S-protein for 10 days. 3) Corresponding epitopes for each expanded TCR clones. After analyzing these results, the paper reports two major findings & claims: A) Individuals having sustained anti-S protein antibody response also have

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We will carefully describe the interpretation of the data with statistical analysis in the revised manuscript.

1. The paper does not provide any information to justify its cell annotation as presented in Fig 2B, 4A. Moreover, in my opinion, it is strange to see that there are two clusters of cells sit on both the left and right side of UMAP in Fig2B but both are annotated as CD4 Tcm and Tem. Also Tfh and Treg belong to a same cluster in Fig 2B but they should have very distinct transcriptomes and should be separated nicely. Therefore I believe the paper can be more convincing if it can present more information and discussion about the basis for its cell annotation.

We apologize for the insufficient explanation and will describe how we performed cell annotation in the revised manuscript.

1. Line 103-104, the paper claims that the Tfh cluster likely comes from cTfh cells. However considering the cells have been cultured/stimulated for 10 days, cTfh cells might lose all Tfh features after such culture. To my best knowledge there is no literature to support the notion that cTfh cells after stimulated in vitro for 10 days (also in the presence of IL2, IL7 and IL15), can still retain a Tfh phenotype after 10 days. It is possible that what actually happens is, instead of having more S-specific cTfh cells before the cell culture, the sustainers' PBMC can create an environment that favors the Tfh cell differentiation (such as express more pro-Tfh cytokines/co-stimulations). Thus after 10-days culture, there are more Tfh-like cells detected in the sustainers. The paper may need to include more evidence to support cTfh cells can retain Tfh features after 10-days' culture.

We thank the reviewer for raising this important point. We will describe the limitation of the strategy. In addition, we will include some data in accordance with the reviewer's recommendation.

1. *It is in my opinion inaccurate to use cell number in Fig4B to determine whether such clone expands or not, given that the cell number can be affected by many factors like the input number, the stimulation quality and the PBMC sample quality. A more proper analysis should be considered by calculating the relative abundance of each TCR clone in total CD4 T cells in each timepoint.*

We will also show the proportion of clonotypes in the revised manuscript.

1. *It is well-appreciated to express each TCR in cell line and to determine the epitopes. However, the author needs to make very sure that this analysis is performed correctly because a large body of conclusions of the paper are based on such epitope analysis. However, I notice something strange (maybe I am wrong) but for example, Table 4 donor #8 clonotype post_6 and _7, these two clonotypes have exactly the same TRAV5 and TRAJ5 usage. Because alpha chain don't have a D region, in theory these clonotypes, if have the same VJ usage, they should have the same alpha chain CDR3 sequences, however, in the table they have very different CDR3a aa sequences. I wish the author could double check their analysis and I apologize in advance if I raise such questions based on wrong knowledge.*

We thank the reviewer for carefully reading our manuscript. Although the two clonotypes, donor #8 clonotype post_6 and _7, have exactly the same TRAV5 and TRAJ5 usage, they have different CDR3a aa sequences due to random nucleotide addition in rearrangement. Likewise, donor #27 clonotype post_1 and donor #13 clonotype post_15 had the same TRAV9-2 and TRAJ17 usage but different CDR3a.